SUMMARY

- Light has wave as well as particle nature.
- A *wavefront* is the locus of points which are in the same state or phase of vibration. Huygen's principle states the method of propagation of wavefront.
- Light propagating as a wavefront. Point source produces spherical wavefront and source at infinity produces plane wavefront.
- Laws of reflection and refraction are proved by Huygens' principle.
- The phenomenon of addition or superposition of two light waves which produces increase in intensity at some points and decrease in intensity at some other points is called *interference* of light.
- If I_1 and I_2 are the intensities of interfering light, then

  $I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$; $I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$
- If interfering lights have equal intensities of I_0 with phase difference ϕ, then

  $I = 4I_0 \cos^2(\phi/2)$

  $I_{max} = 4I_0$ when, $\phi = 0, \pm 2\pi, 4\pi ...,$

  $I_{min} = 0$ when, $\phi = \pm\pi, \pm 3\pi, \pm 5\pi ...,$
- Two light sources are said to be coherent if they produce waves which have same phase or constant phase difference, same frequency or wavelength monochromatic), same waveform and preferably same amplitude.
- Coherent sources are obtained by wavefront division, intensity division and real and virtual images of light source.
- Young's double slit uses wavefront division to obtain coherent sources.
- In Young's double slit experiment, the position of nth bright fringe is, $y_n = n\frac{\lambda D}{d}$
- In Young's double slit experiment, the position of nth dark fringe is, $y_n = \frac{(2n-1)}{2}\frac{\lambda D}{d}$
- In Young's double slit experiment, the equation for bandwidth is, $\beta = \frac{\lambda D}{d}$
- Interference with polychromatic (white) light produces coloured interference fringes.
- Thin films appear coloured due to interference of white light.
- In thins films the equations for constructive and destructive interference for reflected rays are, $2\mu d = (2n-1)\frac{\lambda}{2}$; $2\mu d = n\lambda$ respectively.
- Transmitted light are, $2\mu d = n\lambda$; $2\mu d = (2n-1)\frac{\lambda}{2}$ respectively.
- *Diffraction* is bending of waves around sharp edges into the geometrically shadowed region.
- A spherical wave front undergoes diffraction in Fresnel diffraction.

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